

Subham Das

B.Tech CSE (AI/ML) · 2nd Year · NIMS University, Jaipur

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EDUCATION

B.Tech — Computer Science & Engineering (AI/ML)

2024 – Present

NIMS University, Jaipur

TECHNICAL SKILLS

Languages: Python, C, C++

AI / ML: Machine Learning fundamentals, NEAT (NeuroEvolution), Data preprocessing, NumPy, Pandas, Pygame

Web & Backend: HTML, CSS, REST APIs, Django, React

Cloud & DevOps: AWS (EC2, S3, IAM, RDS, VPC, Boto3), Linux, Bash

Databases: MySQL, MongoDB, PostgreSQL

Tools: Git, GitHub, VS Code, Selenium, BeautifulSoup

Concepts: Data Structures & Algorithms, OOP, OS, Computer Networks, System Design

PROJECTS

AI Certificate Authentication System — Python · AWS · Hashing

2025

- Built a certificate anti-forgery platform that uses image processing to extract certificate data, generates a SHA-256 cryptographic hash per certificate, and stores records in AWS S3 for tamper-proof verification.
- Implemented one-click re-verification by fetching stored hashes from S3 and cross-matching against newly uploaded certificates, eliminating duplicate and forged credentials with zero false-positives in testing.
- Awarded Runner-Up at the NIMS Ideathon & Hackathon 2025, selected from all participating college teams after a live demo and judging panel evaluation.

CHIP-8 Emulator — C / C++

2025

- Built a fully functional CHIP-8 emulator from scratch in C/C++, decoding and executing all 35 standard opcodes with cycle-accurate timing, 4KB memory layout, and 64×32 pixel display rendering.
- Engineered CPU registers (16 × 8-bit), a 16-level call stack, dual countdown timers, and hex keypad input, enabling classic ROM playback including Space Invaders and Tetris.

Data Scraping Pipeline — Python · Selenium · BeautifulSoup · MySQL

2026

- Built automated scrapers using BeautifulSoup (static) and Selenium (JavaScript-rendered pages) to extract structured datasets from multiple target sites, handling pagination and anti-bot measures.
- Cleaned and normalised raw HTML data into CSV, JSON, and SQL formats; loaded into MySQL with indexed schemas to support fast querying and downstream analysis.

Pong AI (NEAT) — Python · NEAT · Pygame

2026

- Trained AI agents to play Pong autonomously using the NEAT (NeuroEvolution of Augmenting Topologies) algorithm, evolving neural network weights and topology over successive generations.
- Designed the fitness function, speciation parameters, and Pygame game loop; agents evolved from random paddle movement to reliable rally-sustaining behaviour within ~30 generations.

RELEVANT COURSEWORK & LAB WORK

- AWS Cloud Practicals — Hands-on labs covering IAM, S3 bucket management, EC2 instance setup (Amazon Linux & Ubuntu), RDS, VPC configuration, and Boto3 scripting
- Operating Systems — Process scheduling (SRTF, Banker's Algorithm), memory management, file systems, deadlocks

- Computer Networks — OSI/TCP-IP model, subnetting, routing protocols, network security
- System Design — Scalability, CAP theorem, load balancing, database design, microservices
- Python Programming — OOP, file handling, data structures (linked lists, stacks, queues, BFS/DFS), NumPy, Pandas
- Probability & Statistics — Binomial & Poisson distributions, normal distribution, hypothesis testing

ACHIEVEMENTS & ACTIVITIES

- Runner-Up — NIMS Ideathon & Hackathon 2025: Presented an AI-based certificate authentication system competing against college-wide participants
- Freelance AI image generation work on Fiverr using Stable Diffusion — delivered custom outputs for international clients, demonstrating applied generative AI skills
- Studying for AWS Certified Cloud Practitioner — completed hands-on labs across EC2, S3, IAM, RDS, VPC, and Boto3 automation scripting

INTERESTS

Machine Learning · Systems Programming · Cloud Computing · Generative AI · Software Development